FlightPlan Bedside Rounds Design Challenge

Principal Investigators: Brianne Reedy, Ryan Moore, and Paul Arichai

The Problem

FlightPlan is a digital tool that organizes a complex inpatient hospitalization into a single trajectory view, analogous to an airplane flight plan. Complex inpatients often have long, eventful hospitalizations, with important context spread across notes, procedures, imaging, conferences, handoffs, and team memory. FlightPlan brings that context together, but the current experience was designed for review conversations away from the bedside and may be too complicated for use at the bedside and with patients and families. This project designs the bedside view that turns FlightPlan into something the team can use during rounds. A simplified bedside view could help when…

- Time is limited, and the team needs the "what changed and what matters now" view immediately.
- Attendings, fellows, residents, nurses, APPs, and consultants may need different levels of detail.
- Patients and families need help understanding the care trajectory.
- The clinical team needs to capture information at the bedside.

Current Stage

FlightPlan is built and in use for review conversations today. It organizes the hospitalization journey around a trajectory:

- **The patient at a glance:** patient/admission summary, current status, review dates, and key diagnoses.
- **The course over time:** location path, respiratory support and acuity changes, procedures, imaging, bedside events, and therapies.
- **Team communication:** annotations, conferences, action items, feedback, documents, and attachments.
- **A shared awareness layer, not an EHR:** FlightPlan does not replace the EHR or generate clinical recommendations; it helps the team understand the patient's course quickly and communicate the plan clearly.

Jumping Off Points for Hackathon Teams

Bring us a fresh, outside perspective. Teams inside healthcare get stuck in well-worn pathways, and you are not bound by them. Interview the clinicians we provide, then create a wireframe for the first bedside rounding version of FlightPlan. **Work from** the existing trajectory view as your starting point, then reimagine what a clinician should see, touch, hide, capture, or hand off during bedside rounds.

The goal is not to redesign the whole application; it is to rethink this one moment of care.

Things to consider, or not!

- What do clinicians need to know in the first 30 seconds at the bedside?
- Which current FlightPlan details are essential, optional, or distracting during rounds?
- What information is safe to show when families are present?
- What should be captured during rounds: decisions, questions, tasks, contingencies, consult needs, family updates, or follow-up?
- What would make clinicians trust this view in a live care workflow?

Why Choose This Project

This is a chance to work on a real clinical workflow where design matters, moving FlightPlan from retrospective review into live bedside care. If you want to help clinicians make sense of complex admissions faster and leave rounds with a better shared plan, choose FlightPlan. This is a **Great fit for:** UX and product design, mobile and tablet interfaces, clinical informatics, and front-end skills. Curiosity about clinical workflows matters more than healthcare experience.

SMART@Home Clinician Data-View Design Challenge

Principal Investigators: Kevin Hommel, Matt Wortman, and Rachel Ramsey

The Problem

SMART@Home (Self-Management Assistance for Recommended Treatment at Home) is a digital health platform built at Cincinnati Children's that helps patients manage chronic and complex conditions at home. SMART@Home is now collecting rich patient data, but on the clinician side it arrives as raw rows in a table that are not practical to use in a visit. This project designs the clinician data view that turns that data into something providers can read at a glance and act on. A purpose-built view could help when...

- Time is short and the clinician needs the "what changed and what matters now" view immediately.
- Different specialists need different levels of detail from the same patient data.
- The team needs to compare a patient's status and responses over time, which is hard today.

Current Stage

Home-monitoring apps are usually one-off builds: one condition each, months of custom development, and little grounding in behavior change, so patients disengage and clinicians get little usable data. SMART@Home solves this:

- **Built on behavioral science:** 20+ years of adherence research built in (positive reinforcement, gamification, reminders).
- **Co-created:** trackers and care logic are co-designed by the patients and clinicians who use them.
- **No software development:** clinicians configure condition-specific trackers in a no-code interface; what took months takes hours.
- **Rapid adaptation:** new trackers deploy in minutes and update instantly from patient feedback.
- **One unified app:** patients manage many conditions (asthma, IBD, migraine, and more) in one app, with data flowing into the Epic EHR.

Jumping Off Points for Hackathon Teams

Bring us a fresh, outside perspective. Teams inside healthcare get stuck in well-worn pathways, and you are not bound by them. Interview the clinicians we provide, then create a wireframe for the SMART@Home clinician data view that reimagines what a clinician should see, compare, and act on when they open a patient's data. Work from the existing tabs (Patients, Trackers, Studies) and per-patient logs as your starting point, then reimagine how that data should be aggregated and displayed.
The goal is not to redesign the whole platform; it is to rethink this one view.

Things to Consider, or Not

- What does a clinician need to see in the first 30 seconds of opening a patient's record?
- Which data points drive treatment decisions, and which are noise?
- How do needs differ across specialties viewing the same data?
- How should the view support verifying adherence, spotting trends, flagging anomalies, and guiding treatment?

Why Choose This Project

SMART@Home is already collecting patient data; the missing piece is a clinician view that turns it into better, faster decisions. Choose this project to shape how clinicians care for children with complex conditions. This is a **Great fit for:** UX and product design, data visualization, clinical informatics, and front-end skills. Curiosity about clinical workflows matters more than healthcare experience.

DISCOVER - Innovation, Simulation, and Collaboration using VR

Principal Investigators: Matt Zackoff, Joe Real, Brad Cruse and Team.

The Problem

Training clinicians for high-stakes pediatric scenarios is hard. Critical events are rare, cannot be rehearsed on real patients, and are difficult to deliver consistently across learners. Traditional simulation helps but is resource-heavy and hard to standardize. Even strong programs struggle to show their impact, because the value of training lives inside the experience and is hard to see from the outside. The people who decide whether to adopt a platform (education directors, nurse leaders, administrators, and potential industry partners) rarely put on a headset, so they need the impact made visible to them. And better-prepared clinicians matter: every scenario a trainee can safely practice in a digital twin is one they do not face for the first time on a real child.

Current Stage

DISCOVER is a working digital-twin platform. Scenarios run live and adapt to participant actions, incorporating clinical contexts, pediatric patient avatars, and the full range of equipment in real time. Core interactions are built, the platform captures rich data including eye-tracking, and an AI-assisted version is in testing. Clean CCHMC-owned IP, no federal funding entanglements. The clinical engine is mature; the next step is making its value visible to the people who decide whether to adopt it.

Jumping Off Points for Hackathon Teams

- Outcomes dashboard (primary): visualize how the simulations affect learner performance and clinical outcomes, drawing on data the team has begun to assemble
- Learner “report card” (primary): show how individuals performed within a scenario, tying education tightly to results
- Industry-partner view: surface the rich interaction data DISCOVER already captures (workflows, clinical interactions, eye-gaze) for human-factors and usability applications
- Unified data visualization layer for cohort and individual performance, suitable for program review or partner-facing reporting
- AI-assisted recommendation that suggests the next scenario based on a learner’s history

Things to Consider, or Not!

- The realistic one-day target is a self-contained prototype with sample data, demoable on its own without VR engine integration
- You do not need to build in VR or touch the 3D environment. The opportunity is the data and dashboard layer that sits alongside the platform
- These are new surfaces, so there is no existing design to match or preserve. Bring your own thinking

Why Choose This Project

You will work on a VR platform and build the piece that makes its value visible, the surface that helps a real platform reach clinicians and partners. The work is high-impact and high-visibility: it directly strengthens how the platform reaches the people who train the clinicians who care for kids. The PI and dev team are engaged and available the morning of the event (~1 to 2 hours) to brief teams and answer questions.

Parent Resource Hub (PRH)
Principal Investigator: Shari Wade, PhD

The Problem

Over many years, Dr. Wade and her team have developed and validated a portfolio of internet-based interventions that help families navigate child development, behavior, family functioning, and recovery from pediatric brain injury. These programs work, but they live as separate course-style offerings rather than one cohesive product. For a parent, the experience can feel fragmented: finding the right program, knowing where to start, and staying engaged are all hard. And when support is hard to reach, families disengage when they most need help, so a program that is hard to find or start often goes unused even when it works.

Current Stage

The Parent Resource Hub (PRH) is a live platform (parentresourcehub.org) that brings together several of Dr. Wade's validated programs as online courses accessible from any device. The programs are grounded in published research and real-world testing, including responsive parenting interventions and goal-oriented family support for children with acquired brain injury. The content and structure are currently organized around individual courses rather than a personalized, partner-ready experience. The clinical foundation is in place and proven; the next step is the experience layer that makes it cohesive, personal, and partner-ready.

Jumping Off Points for Hackathon Teams

- Information architecture and onboarding, so a parent can quickly find the right program and know where to begin
- A personalization layer that recommends programs based on a family's situation
- An AI-assisted guide or triage experience that helps parents navigate content and answers common questions
- A partner-facing view (landing experience, white-label concept, or engagement dashboard) that demonstrates value to a health system or payer
- Engagement and re-engagement features: reminders, nudges, progress visualization, session prompts
- Modern visual design, accessibility, and mobile experience; a simpler sign-in and first-session flow

Things to Consider, or Not!

- The validated clinical course content stays as is. The opportunity is everything around it: how families find, start, and stay with the right program
- A self-contained prototype with sample content is a great one-day target; you do not need to integrate with the live platform
- Pick the angle that excites your team. You do not need to tackle all of these

Why Choose This Project

You will work on a live platform built on years of validated research and shape how real families experience it, the difference between proven content sitting in separate courses and proven content that actually reaches the families who need it. The work is human, visible, and consumer-facing, with room for strong design and product thinking. Dr. Wade's team is engaged and available the morning of the event to brief teams and answer questions.

Digital Health Hackathon 2026 | Technology One-Pager

Digitally-Enabled Adherence Platform (DEAP)

Principal Investigator: Avani Modi, PhD; David Hooper, MD; Charles Varnell, MD

The Problem

Not taking medication as prescribed is one of the biggest hidden drivers of poor health outcomes, and it hits young transplant patients especially hard. For an adolescent with a kidney transplant, missed immunosuppressant doses can lead to organ rejection, hospitalization, and death. At Cincinnati Children's, nearly half of kidney transplant patients and caregivers report at least one barrier to taking their medication, and those who do are nearly three times more likely to experience acute rejection. For a transplant family, a missed dose is not a minor lapse; it can mean rejection of a donated organ or a life-threatening event. Yet barriers are rarely assessed in a structured way, and there is no standard method to deliver proven strategies to help patients overcome them between visits.

Current Stage

DEAP is built on the team's clinic-based expertise, which was shown to reduce kidney transplant rejection episodes while actually saving money. DEAP is software-only, with no expensive proprietary hardware, connecting instead to the communication tools patients already use, and it personalizes the timing and content of messages to each patient's barriers, cognitive load, language, and schedule. The clinical foundation is proven; the opportunity now is the experience that carries it to families at scale.

Jumping Off Points for Hackathon Teams

The opportunity is the patient-and-family-facing experience layer on top of the evidence-based clinical content. Possible starting points:

- A barrier-assessment and onboarding experience that helps a patient or caregiver quickly identify what is getting in the way
- A personalization layer that adapts message timing and content to a patient's situation and preferences
- Engagement and re-engagement features: reminders, nudges, streaks, progress visualization
- A patient-facing dashboard that makes adherence progress easy to see
- A clinician-facing view showing which patients need attention and which interventions work
- Modern visual design, accessibility, and mobile experience

Things to Consider, or Not!

- The evidence-based clinical content and adherence strategies stay as is. The opportunity is the experience layer that delivers them
- A self-contained prototype with sample data is a strong one-day target; no integration with live patient data is needed
- Choose the surface that fits your team's strengths. You do not need to build all of it

Why Choose This Project

You will work on a platform built on a clinic-proven program that reduced organ rejection while saving money, and build the experience that helps it reach families every day. The problem is enormous (about half of patients with chronic conditions do not take medications as prescribed, at a cost of roughly $528 billion and 125,000 lives a year), and the surface is squarely consumer-facing: messaging, personalization, dashboards, and engagement. The team is engaged and available the morning of the event to brief teams and answer questions.